Multi-Class Classification of Jailbreak Attack Prompts into Safety Harm Categories

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Abstract

Jailbreak prompts are adversarial instructions aiming at bypassing the safety mechanisms of large language models (LLMs), posing significant challenges for their safe and trustworthy deployment. This work frames jailbreak harm detection as a 16-class text-classification problem on the recently released JailBreakV-28K dataset of 28,000 prompts. We develop an end-to-end pipeline including exploratory data analysis, text preprocessing, stratified data splitting, multiple text representations, hyperparameter tuning, model evaluation, error analysis, soft-voting ensembling, and input-ablation analysis. We experiment with classical machine-learning models, recurrent neural networks and their bidirectional variants, and BERT Base . The best-performing individual model, TF–IDF with Logistic Regression, achieved 0.8886 accuracy and 0.8724 Macro-F1 on the test set. A soft-voting ensemble of Logistic Regression, Bidirectional GRU, and GRU models improved the results to 0.8898 accuracy and 0.8733 Macro-F1. Our ablation study demonstrated that the use of the underlying redteam query as input increased the validation Macro-F1 from 0.8833 to 0.9910, illustrating that the explicit intent information included in the redteam query helps improve the model performance. Overall, our results highlight the competitiveness of lightweight lexical models for the jailbreak classification task while raising questions about the impact of class imbalance, use of repeated templates, and adversarial formatting on the model performance.

1 Introduction

Large language models (LLMs) are widely useful for search, writing, programming, education, and decision support, but their broad capabilities also pose safety risks. Modern LLMs employ refusal mechanisms, content filtering, and alignment techniques, but jailbreak prompts are designed to bypass these protections through strategies such as role-play and formatting manipulation. This work examines jailbreak harm categorization as a 16-class text-classification problem using the jailbreak query field from JailBreakV28K (1). The task includes categories such as Malware, Fraud, Violence, Privacy Violation, Hate Speech, Illegal Activity, and Health Consultation. The dataset is challenging due to class imbalance and repeated jailbreak templates. We compare classical machine-learning models, recurrent neural networks, bidirectional recurrent models, and BERT Base using TF–IDF, Word2Vec, GloVe, and WordPiece representations. All models are tuned using validation Macro-F1 and evaluated on a held-out test set. The study investigates which representation and architecture performs best, whether gated bidirectional recurrent models improve performance, and whether ensemble prediction provides an additional benefit. The results suggest that TF–IDF with Logistic Regression is the strongest individual model and that a soft-voting ensemble provides a small further improvement.

2 Related Work

Text classification has traditionally used sparse lexical representations and linear or probabilistic models. TF–IDF gives higher weight to terms that are frequent in a document but less common outside of it, making it effective when class is defined by distinctive words or phrases (2). Logistic Regression is especially suited for high-dimensional sparse inputs because it learns linear decision boundaries efficiently, whereas Multinomial Naive Bayes offers a probabilistic baseline that is computationally cheap. Random Forest offers a nonlinear ensemble alternative, but is generally less natural for incredibly sparse text spaces than linear models.

Distributed word representations were introduced to encode semantic similarity in dense vector spaces. Word2Vec (3) learns embeddings from local context by optimizing objectives such as skip-gram, whereas GloVe (4) derives vectors from global co-occurrence statistics. Unlike TF–IDF, which uses sparse lexical dimensions to represent a document, embedding-based neural models can capture relationships between semantically related words and process text sequentially.

While vanilla SimpleRNN designs can have trouble with long-range dependencies, recurrent neural networks are built to accommodate sequential inputs. LSTM networks (5) introduce gated memory cells, while GRUs (6) have a simpler gating mechanism that oftentimes offers comparable performance with fewer parameters. Bidirectional recurrent models process the same sequence in both directions and the final representation incorporates context from both sides of each position, which is useful for jailbreak prompts since harmful intent may be expressed through combinations of instructions distributed across a prompt.

Transformer models offer an alternative to contextual representation. To create contextual token representations, BERT (7) uses deep bidirectional self-attention and masked-language-model pretraining. Fine-tuned Transformer models have achieved high accuracy on many classification tasks, but its benefit depends on the data distribution, sequence lengths, label semantics, training configurations, and the amount of task-specific data provided.

Jailbreak safety research has increasingly focused on adversarial prompt creation and robustness testing. JailBreakV-28K (1) is a large benchmark with multiple jailbreak formats and policy labels. This project focuses on the supervised classification side of the problem: given a jailbreak prompt, does any choice of representation and architecture choices allow us to recover the associated harm category reliably?

3 Methodology

3.1 Dataset

The experiments make use of JailBreakV-28K, a dataset that has 28000 english jailbreak prompts categorized into 16 safety-policy buckets. Every record stores the adversarial `jailbreak_query`, its underlying `redteam_query`, a policy label, attack-format details, and further metadata. The main experiment only uses the `jailbreak_query` as input and `policy` as the prediction task.

The class distribution is moderately imbalanced. Malware is the largest category with 3828 samples, followed by Illegal Activity with 3404 and Fraud with 2916. On the other hand of the distribution, Health Consultation has 524 samples and Child Abuse Content has 536 samples. Since large classes could dominate the overall accuracy, Macro-F1 is applied as the main model-selection metric so that every class has equal say in the score. The largest-to-smallest class ratio is 7.31, which proves that accuracy alone would not be an adequate evaluation for minority-class behavior.

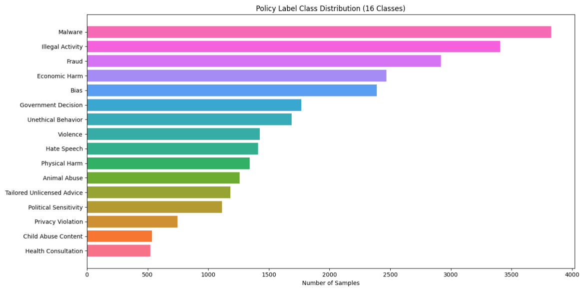


Figure 1: Distribution of the 16 policy classes in JailBreakV-28K.

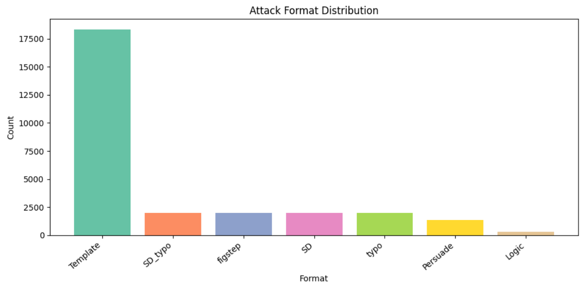


Figure 2: Distribution of the seven jailbreak attack formats. Template attacks dominate the dataset with 18,336 samples.

3.2 Exploratory Data Analysis

Exploratory data analysis is applied before model training to understand the data characteristics and distribution. The prompt length varies significantly. The mean `jailbreak_query` length is about 109.6 words, and the median is 75 words. The distribution is right-skewed, containing both shorter direct queries and longer template-based attacks. The query length varies from several words to over a hundred words.

Moreover, multiple attack formats are included in the dataset. Template-based attacks are the most common, and other formats such as typo and persuasion-based attacks also exist. It is important to note that the classifier might capture the semantic content of harmful queries and attack formats simultaneously. There is a significant number of repeated `jailbreak_query` strings, which indicates that the lexical patterns might be predictive in a

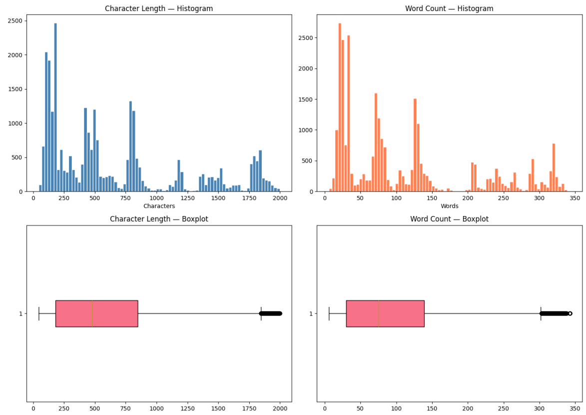


Figure 3: Distribution of jailbreak-query lengths.

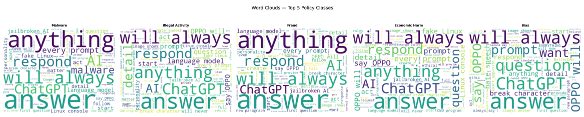


Figure 4: Frequent terms in representative policy categories.

random stratified split.

Seven attack formats are included in the dataset: Template (18336), SD_typo (2000), figstep (2000), SD (2000), typo (2000), Persuade (1368), Logic (296). Thus, template based attacks are predominant, which might introduce significant repetition in the model training data.

There were no completely repeated rows in the dataset. However, there were 19359 repeated `jailbreak_query` strings. It might be notable that the same adversarial prompt wrapper could have different adversarial intents or policy labels attached, and these differences could impact the model performance. Therefore, a random stratified split would contain different prompt templates, which could lead to different model training samples.

3.3 Preprocessing

The preprocessing strategy is deliberately conservative. All text is lowercased, repeated whitespace is normalized, and any leading or trailing spaces are stripped. Labels are encoded as integer class identifiers. This minimal cleaning process preserves potentially informative words, instruction markers, and jailbreak-style phrasing that could be lost in aggressive normalization.

A stopword-removal variant is also tested as a small preprocessing probe. Only a marginal improvement is found when stopwords are removed, so the main experiments use minimally processed text. The motivation for this choice is that function words and repeated instructional structures might be integral to the attack templates and thus useful to the classifier. In a three-fold TF–IDF/Logistic Regression probe, retaining stopwords achieves 0.7351 Macro-F1, whereas removing them achieves 0.7452, a small difference of 0.0101. The main pipeline nevertheless retains conservative preprocessing to preserve the original adversarial phrasing.

3.4 Train, Validation, and Test Split

The data was partitioned using a reproducible stratified split in order to preserve approximately the same class proportions in each subset. The final split comprised of 19,611 training (70%), 4,189 validation (15%) and 4,200 test (15%) samples. The validation set was used to manually tune the hyperparameters whereas the test set was untouched until the best configuration of each model family had been selected.

Same random seed was used in all data partitioning to ensure reproducibility. As the dataset included repeated templates, stratification preserved label balance but not template independence. This problem will be discussed further later as an important limitation.

3.5 Text Representations

TF–IDF. The classical machine-learning models use TF–IDF features extracted from both unigrams and bigrams. The vocabulary size is capped at 20,000 features, with `min_df=2`, `max_df=0.95` and sublinear term-frequency. The goal is to capture short instruction patterns not captured by individual words via bigrams.

Word2Vec. A 200 dimensional Word2Vec skip-gram is pretrained utilizing the training corpus. The context window was 5, minimum count was 2, and it was trained for 10 epochs. Word2Vec embeddings were applied using the unidirectional recurrent models. Training the embeddings on the project corpus means that the representation was able to adapt to recurring jailbreak vocabulary and attack-related terminology. The Keras vocabulary covers 6,432 tokens, of which 5,893 match Word2Vec, representing 91.6% vocabulary coverage.

GloVe. Pretrained 200-dimensional GloVe 6B vectors are used for the bidirectional recurrent models. GloVe provides a broader semantic initialization drawn from a large external corpus and thus acts as a complementary representation to the Word2Vec embeddings specific to the project. It

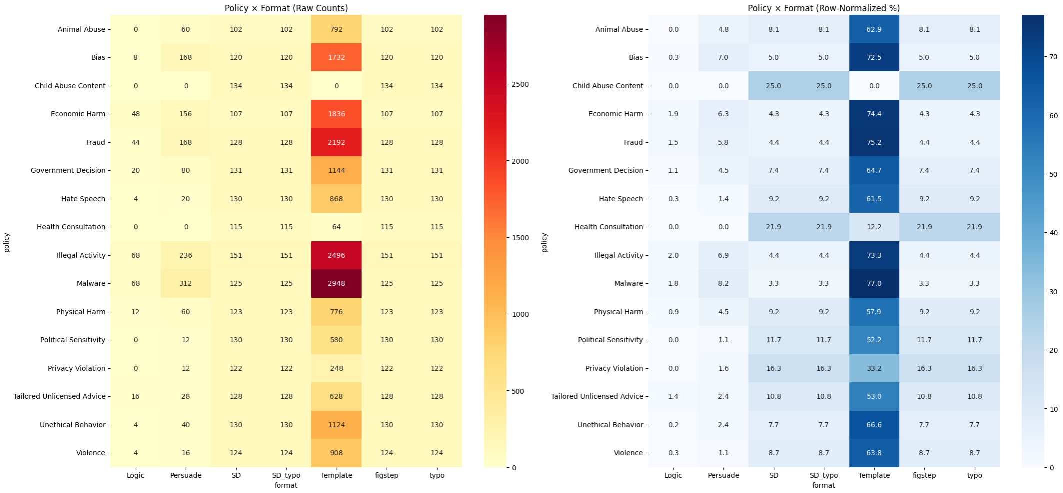


Figure 5: Relationship between policy labels and jailbreak attack formats.

matches 5,977 of the 6,432 Keras vocabulary entries, representing 92.9% coverage.

BERT. BERT Base makes use of the `bert-base-uncased` tokenizer and pretrained model. Inputs are padded/truncated to a maximum length of 128 WordPiece tokens. The final representation undergoes fine-tuning for the 16-class classification objective.

3.6 Models

Three classical machine-learning models are trained which are Logistic Regression, Multinomial Naive Bayes, and Random Forest. The former serves to provide a strong linear baseline for TF–IDF features, the latter provides a probabilistic baseline, and the third provides a nonlinear ensemble comparison.

The recurrent neural models are SimpleRNN, GRU, and LSTM. Their bidirectional counterparts; Bidirectional SimpleRNN, Bidirectional GRU, and Bidirectional LSTM are trained separately. BERT Base is used as the Transformer baseline. In total, 10 required individual models are evaluated.

3.7 Hyperparameter Tuning

Each model is manually tuned with three distinct sets of hyperparameters, as per the project specification. The best configuration is selected according to the highest Validation Macro-F1 score. Class weighting is used in order to reduce frequent-class dominance in neural architectures, and early stopping is applied when necessary to prevent overfitting.

The preferred configuration for every model is presented in Table 1. Logistic Regression achieves

Model	Best configuration	Val. Acc.	Macro-F1
Logistic Regression	$C = 10$, balanced	0.8950	0.8819
Bi-GRU	64 units, lr=10^{-3}	0.8935	0.8800
GRU	64 units, lr=10^{-3}	0.8892	0.8745
Bi-LSTM	128 units, lr=10^{-3}	0.8852	0.8736
LSTM	64 units, lr=10^{-3}	0.8794	0.8674
Random Forest	100 trees, no depth limit	0.8783	0.8696
Bi-SimpleRNN	64 units, lr=10^{-3}	0.7367	0.7246
SimpleRNN	64 units, lr=10^{-3}	0.6149	0.5710
Naive Bayes	$\alpha = 0.1$	0.5791	0.5933
BERT Base	lr=2×10^{-5}, bs=16	0.5682	0.6220

Table 1: Best validation configuration for each model, ordered by validation accuracy from highest to lowest.

the best performance when using $C = 10$ and balanced class weights with TF–IDF as the vectorizer, while 64-unit GRU and LSTM configurations are the most efficient for the recurrent models, and Bidirectional GRU provides the best neural validation Macro-F1.

4 Results

4.1 Overall Test Performance

After selecting the best configuration of each model using the validation set, all ten models are evaluated on the held-out test set. Table 3 displays the accuracy, Macro-F1, and weighted-F1 of all tested models. It is visible that Logistic regression has the greatest single-model Macro-F1 of 0.8724 and it has an accuracy of 0.8886. Moreover, Bidirectional GRU has a Macro-F1 of 0.8691 and a GRU score of 0.8644 which makes this model come second to logistic regression.

Therefore, three different groups are displayed in the ranking. Relatively good results are obtained by Random Forest, LSTM, Bidirectional LSTM, GRU, Bidirectional GRU, and Logistic Regression. Bidirectional SimpleRNN performs much worse

Model	Configuration	Val. Acc.	Macro-F1	Weighted-F1
Logistic Regression	$C = 10$, balanced	0.8950	0.8819	0.9208
Bi-GRU	64 units, lr=10^{-3}	0.8935	0.8800	0.9186
GRU	64 units, lr=10^{-3}	0.8892	0.8745	0.9150
Bi-GRU	128 units, lr=10^{-3}	0.8880	0.8765	0.9170
Bi-LSTM	128 units, lr=10^{-3}	0.8852	0.8736	0.9109
GRU	128 units, lr=10^{-3}	0.8821	0.8626	0.9046
LSTM	64 units, lr=10^{-3}	0.8794	0.8674	0.9068
Random Forest	100 trees, depth=None	0.8783	0.8696	0.8858
Bi-LSTM	64 units, lr=10^{-3}	0.8775	0.8643	0.9024
Random Forest	300 trees, depth=None, balanced	0.8744	0.8610	0.8994
LSTM	128 units, lr=10^{-3}	0.8661	0.8542	0.8966
Random Forest	300 trees, depth=30	0.8518	0.8424	0.8597
Bi-GRU	128 units, lr=5×10^{-4}, dropout=.3	0.8496	0.8377	0.8832
GRU	128 units, lr=5×10^{-4}, dropout=.3	0.8494	0.8396	0.8830
Logistic Regression	$C = 1.0$, L2	0.8133	0.7808	0.8175
Bi-LSTM	128 units, lr=5×10^{-4}, dropout=.3	0.8109	0.7980	0.8392
LSTM	128 units, lr=5×10^{-4}, dropout=.3	0.7787	0.7344	0.8151
Bi-SimpleRNN	64 units, lr=10^{-3}	0.7367	0.7246	0.7709
Bi-SimpleRNN	128 units, lr=10^{-3}	0.6441	0.6412	0.6877
SimpleRNN	64 units, lr=10^{-3}	0.6149	0.5710	0.6630
SimpleRNN	128 units, lr=5×10^{-4}, dropout=.3	0.6009	0.5713	0.6307
BERT Base	lr=5×10^{-5}, bs=16	0.5880	0.6202	0.6258
BERT Base	lr=3×10^{-5}, bs=32	0.5863	0.6157	0.6235
Bi-SimpleRNN	128 units, lr=5×10^{-4}, dropout=.3	0.5827	0.5937	0.6209
Naive Bayes	$\alpha = 0.1$	0.5791	0.5933	0.6346
BERT Base	lr=2×10^{-5}, bs=16	0.5682	0.6220	0.6069
Naive Bayes	$\alpha = 1.0$	0.5393	0.5125	0.5458
Logistic Regression	$C = 0.1$, L2	0.4832	0.3696	0.4382
SimpleRNN	128 units, lr=10^{-3}	0.4605	0.4727	0.5305
Naive Bayes	$\alpha = 5.0$	0.3039	0.1893	0.2437

Table 2: Complete manual hyperparameter tuning results, ordered by validation accuracy from highest to lowest.

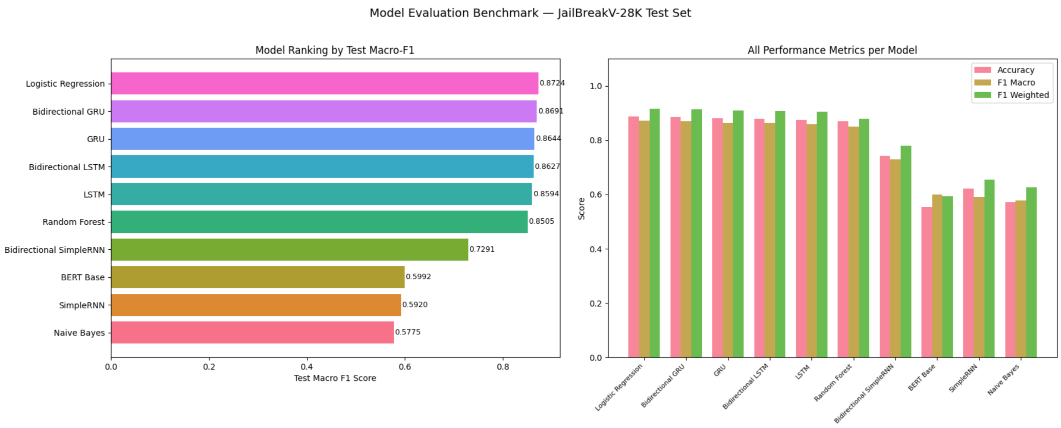


Figure 6: Comparison of accuracy, Macro-F1, and weighted-F1 across models.

Model	Acc.	Macro-F1	Wtd.-F1
Logistic Regression	**0.8886**	**0.8724**	**0.9172**
Bidirectional GRU	0.8864	0.8691	0.9145
GRU	0.8807	0.8644	0.9101
Bidirectional LSTM	0.8783	0.8627	0.9070
LSTM	0.8750	0.8594	0.9048
Random Forest	0.8702	0.8505	0.8779
Bidirectional SimpleRNN	0.7429	0.7291	0.7791
SimpleRNN	0.6226	0.5920	0.6548
Naive Bayes	0.5719	0.5775	0.6255
BERT Base	0.5543	0.5992	0.5926

Table 3: Held-out test performance of all required models, ordered by test accuracy from highest to lowest.

than the gated recurrent architectures. SimpleRNN, BERT Base, and Naive Bayes form the bottom performing group under the current configuration.

4.2 Best and Worst Models

The best individual model is class-balanced Logistic Regression with TF–IDF. The result is notable since it slightly outperforms more complex neural systems. It is likely because the policy labels in JailBreakV-28K are connected to strong lexical clues, repeated attack templates and category-specific statements. Unigram and bigram TF–IDF features can represent these signals directly and Logistic Regression can sort them efficiently in a high-dimensional sparse feature space.

Bidirectional GRU is the strongest neural model. It benefits over the unidirectional GRU by processing the prompt in both directions and achieves a small improvement in Macro-F1. LSTM and Bidirectional LSTM also show similar patterns, although the benefit is more modest. These results suggest that gated recurrent memory is helpful for the task, but bidirectionality provides a minor sec-

Class	P	R	F1	Support
Animal Abuse	0.98	0.90	0.94	189
Bias	0.99	0.93	0.96	358
Child Abuse Content	0.16	0.95	0.28	80
Economic Harm	0.99	0.91	0.95	370
Fraud	0.99	0.93	0.96	437
Government Decision	0.96	0.86	0.91	265
Hate Speech	0.94	0.87	0.91	212
Health Consultation	1.00	0.72	0.84	79
Illegal Activity	0.98	0.89	0.93	511
Malware	0.99	0.94	0.97	574
Physical Harm	0.96	0.85	0.90	201
Political Sensitivity	0.91	0.76	0.83	167
Privacy Violation	0.93	0.70	0.80	112
Tailored Unlicensed Advice	0.99	0.88	0.93	178
Unethical Behavior	0.98	0.87	0.92	253
Violence	0.98	0.90	0.94	214

Table 4: Classification report for the best Logistic Regression model.

ondary benefit.

The lowest Macro-F1 of 0.5775 is attained with Naive Bayes. For adversarial prompts with structured multiword instructions, its conditional-independence assumptions are rigid. BERT Base also performs far below the best classical and recurrent models. It does not necessarily mean that Transformer models are generally inappropriate for jailbreak classification; it suggests that the current data distribution and fine-tuning configuration better favor lexical and recurrent representations.

4.3 Per-Class Performance

The complete classification report of the best Logistic Regression model is presented in Table 4. Achieved precision and recall for the majority of classes are high. Malware obtains an F1-score equal to 0.97, while Bias and Fraud both reach 0.96. Classes including Animal Abuse, Violence, Economic Harm, Illegal Activity, Tailored Unlicensed Advice, Unethical Behavior, and Government Decision also exhibit high precision and recall.

The only exception is Child Abuse Content, for which the classifier attains a very high recall but a low precision, which implies that it predicts this minority label too often. Such an outcome indicates that certain lexical or template-level features associated with this class are also present in prompts corresponding to other categories. Similarly, Health Consultation, Privacy Violation, and Political Sensitivity also attain lower F1-scores than the best classes, which may be due to their smaller support or higher semantic overlap.

4.4 Confusion Matrix Analysis

The Logistic Regression confusion matrix offers a deeper insight into the error distribution. The model succeeds in classifying most examples in the larger categories, but several prompts from other policies are wrongly labeled as Child Abuse Content. The error pattern is compatible with the combination of class imbalance, repeated templates, and overlapping lexical cues.

Figure 7 maintains the confusion matrices together and ranks them according to held-out test accuracy. The ensemble, Logistic Regression, and Bidirectional GRU exhibit the strongest diagonal concentration, followed by Bidirectional LSTM and Random Forest. SimpleRNN, Naive Bayes, and BERT Base reveal wider cross-class confusions. This ordered grid allows to compare the performance differences more easily while holding the compact page layout.

4.5 Error Analysis

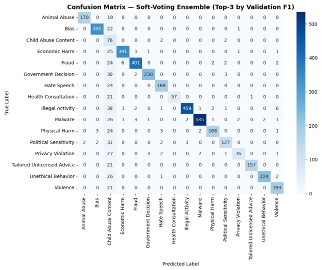
The highest performing model, Logistic regression, incorrectly classifies 468 test cases where total test cases are 4,200, which gives us the error rate of 11.14%. The most common error is Illegal Activity predicted as Child Abuse Content (42 cases). Other confusions into Child Abuse Content are Government Decision (30), Privacy Violation (29), Political Sensitivity (29), Malware (29), Unethical Behavior (27), Economic Harm (25), Physical Harm (24), Fraud (23), and Hate Speech (23).

The associated prompt excerpts suggest that many of these errors share similar template wording, with generic instructions for the model to complete numbered lists or give steps for an activity. This supports the EDA finding that repeated jailbreak wrappers can dominate the lexical signal and cause prompts with different underlying harms to be similar to the classifier.

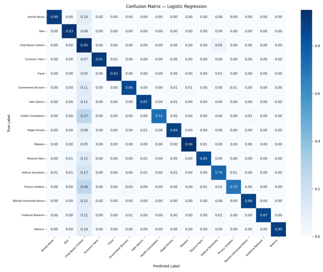
4.6 Soft-Voting Ensemble

A soft-voting ensemble is formed from the three best validation-selected models: Logistic Regression, Bidirectional GRU, and GRU. Their class-probability outputs are averaged to select the final label. The ensemble achieves 0.8898 test accuracy and 0.8733 Macro-F1, slightly outperforming the best model.

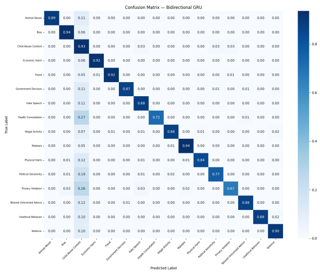
The improvement is small but significant because the constituent systems use different representations. Logistic Regression exploits sparse



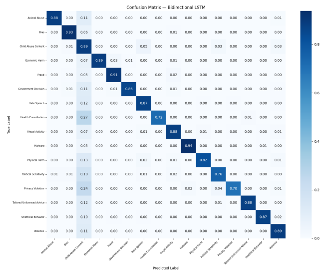
(a) Ensemble — Accuracy: 0.8898



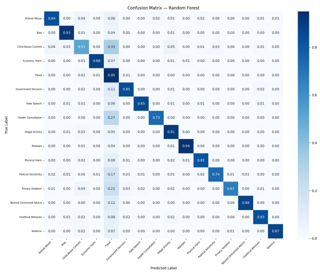
(b) Logistic Regression — Accuracy: 0.8886



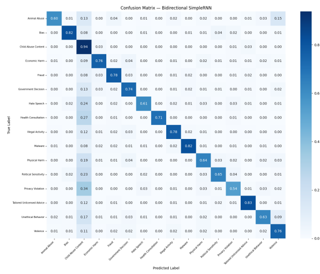
(c) Bidirectional GRU — Accuracy: 0.8864



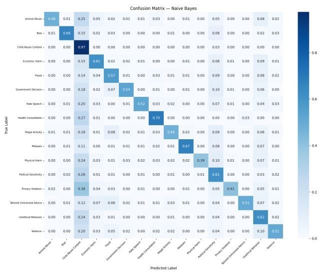
(d) Bidirectional LSTM — Accuracy: 0.8783



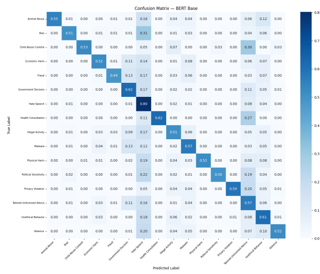
(e) Random Forest — Accuracy: 0.8702



(f) SimpleRNN — Accuracy: 0.6226



(g) Naive Bayes — Accuracy: 0.5719



(h) BERT Base — Accuracy: 0.5543

Figure 7: Confusion matrices ordered by held-out test accuracy from highest to lowest: soft-voting ensemble, Logistic Regression, Bidirectional GRU, Bidirectional LSTM, Random Forest, SimpleRNN, Multinomial Naive Bayes, and BERT Base.

lexical features, while GRU-based models learn sequential representations. Their errors are therefore not entirely identical. Combining them allows the ensemble to benefit from complementary evidence without introducing a substantially more complex meta-model.

4.7 Input-Ablation Study

We conduct an additional experiment that examines the impact of including the `redteam_query` as part of the main `jailbreak_query`. Concatenating the two fields improves the validation Macro-F1 of Logistic Regression from 0.8833 to 0.9910 which is a much higher improvement than achieved by any of the architectural or ensemble changes.

We are careful to interpret this result with caution as the `redteam_query` provides a cleaner description of the original harmful intent that the jailbreak prompt attempts to hide since the policy label is closely related to that underlying intent, the field might reveal some target-related information that would not be available in a real deployment scenario. As a result, we report this as an ablation study rather than incorporating it as the primary model input.

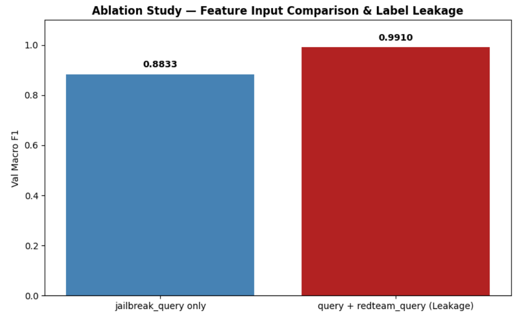


Figure 8: Validation Macro-F1 for jailbreak-query-only input (0.8833) versus jailbreak query plus red-team intent (0.9910). The near-perfect combined-input result is treated as suspected label leakage.

5 Discussion

The results of this work confirm that model complexity does not necessarily improve classification performance on JailBreakV-28K. TF–IDF with Logistic Regression slightly outperforms the best recurrent models and vastly outperforms BERT Base under the current training conditions, ranking as the best individual model with 88.86% accuracy and 0.8724 Macro-F1. Notably consistent with the exploratory analysis: this dataset contains strong lexical regularities and repeated jailbreak templates, wrapped around common wrapper phrases ("will", "always", "anything", "respond") that sparse n-gram features can capture efficiently, without requiring any real semantic reasoning.

Given that BERT is thought to be able to capture stronger contextual connections than either TF–IDF or the recurrent networks, its poor performance (0.5543 accuracy, 0.5992 Macro-F1, and 0.5926 weighted-F1) may be the most unexpected outcome. BERT's advantage seems to disappear when label prediction relies heavily on repeated lexical cues. This matches Mosbach et al.'s (8) account of fine-tuning instability: excessively aggressive learning rates lead to representation collapse in the pretrained lower layers, and the optimizer favors degenerate, near-majority-class predictions. The subword fragmentation of adversarial obfuscation, the use of the pooler output rather than the [CLS] token itself, and early stopping due to a stagnant validation loss were probably additional contributing factors. Sun et al. (9) and Mosbach et al. (8) show that this can be alleviated with AdamW, a warmup schedule, and a 2×10^{-5} learning rate, which would presumably push BERT well above its current rank. It also may have been affected by sequence truncation, class imbalance, and the relatively small set of manual configurations tested here.

The recurrent experiments do offer some insight into sequence models. GRU and LSTM massively outperforming SimpleRNN (0.8644 and 0.8594 Macro-F1 vs. 0.5920) confirm the value of gated memory over long ranges. Bidirectional GRU achieves the best neural result (0.8691 Macro-F1), hinting that information late in an adversarial prompt provides useful context for earlier content, and vice versa, though the bidirectional benefit is smaller than the gain from gated over non-gated models.

The soft-voting ensemble of Logistic Regression, BiGRU, and GRU only achieves a slight improvement in results: 0.8724 $\rightarrow$ 0.8733 Macro-F1, suggesting that lexical and recurrent representations capture mostly overlapping signals. Finally, the ablation study's near-perfect score when `redteam_query` is concatenated with the jailbreak prompt is a caution against interpreting huge jumps in safety-classification performance as improvements in real-world performance: the field reveals a clean version of harmful intent that would never be available to a real guardrail classifier, so such leaps in accuracy should always be double-checked for leakage.

6 Conclusion

This study compares ten models across TF–IDF, Word2Vec, GloVe, and BERT-based representations to give a comprehensive multi-class classification pipeline for jailbreak harm categorization on the 28,000-sample JailBreakV-28K dataset. The strongest individual model was Class-balanced Logistic Regression with TF–IDF (0.8886 accuracy, 0.8724 Macro-F1), followed by Bidirectional GRU, which outperformed neural models (0.8691 Macro-F1), and Naive Bayes. Results were enhanced to 0.8898 accuracy and 0.8733 Macro-F1 using an ensemble of Logistic Regression, BiGRU, and GRU with soft voting. The results indicate that when jailbreak categories have strong word-level signals, basic lexical models can compete successfully. However, performance should be viewed with the warning that the dataset's repetitive templates may limit generalization to innovative jailbreak tactics.

Limitations

The most important limitation is the abundance of similar or structurally related jailbreak-query templates. A random stratified split can put structurally similar prompts in both training and test sets, which may overestimate generalization to completely unseen attack styles. A stronger future evaluation should therefore use template-disjoint, attack-family-disjoint or group-aware splitting. A second limitation is class imbalance. While Macro-F1, class weighting and stratified splitting reduces its effect, the smallest categories contain significantly fewer examples than Malware or Illegal Activity. This is evident in the per-class results, especially for Child Abuse Content, Privacy Violation, Political Sensitivity and Health Consultation.

A third limitation concerns transformer tuning. BERT Base is evaluated over three manual configs as demanded by the project, but a broader search over learning rate, batch size, seq length, layer freezing, warm up and training duration should be able to increase the performance. The BERT result should therefore be seen as the result of the implemented training scenario instead of an upper bound for transformer-based classification.

Finally, the experiments are conducted on only one benchmark. Performance on JailBreakV-28K does not guarantee robustness to new jailbreak techniques, other domains or adversarial inputs that appeared after the dataset was collected. An external evaluation on another jailbreak benchmark would have given stronger evidence of generalization.

Ethical Considerations

There are malicious and hostile prompts in the dataset. Every experiment in this project is carried out for safety analysis, model evaluation, and defense classification. The resulting classifier is meant to help safety review rather than encourage bad conduct, and the report does not replicate needless harmful cases. Model forecasts should not be used as a total substitute for human or policy-level oversight in actual deployment, but rather as one part of a larger safety system.

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